1. Known type masses and timing

There are N consumers, two inside goods $j \in \{1, 2\}$ owned by a monopolist, and an outside option 0. Types are $r \in \{H, L\}$, with known, fixed probability masses

$$\omega_H = \lambda, \qquad \omega_L = 1 - \lambda, \qquad 0 \leq \lambda \leq 1, \qquad \omega_H + \omega_L = 1.$$

Individual type is unobserved in the main common-price problem. The firm chooses its entire price menu before purchases occur, using these known masses. There is no learning, estimation, posterior, or subsequent belief-based repricing. A terminal menu may depend on the purchased product because that product determines the next period's variety utility; this is a preannounced pricing rule. Commitment makes this timing explicit.

Consumers are myopic, while the firm discounts second-period profits by δ. Consumers begin with no prior purchase. First-period utility has no variety bonus; variety utility begins in period two. An outside purchase in period one exits permanently. In period two, h denotes the inside product purchased in period one. All prices are net transaction prices in $[\underline{p}, \overline{p}] = [1, 8]$.

2. Demand: every differentiation step

At a history h, define

$$V_{rj}(h; p) = \beta_j + \gamma_r \mathbb{1}\{j \neq h\} + \alpha p_j, \qquad \alpha < 0, \tag{2.1}$$

$$Z_r(h; p) = 1 + \sum_{k=1}^{2} e^{V_{rk}(h;p)}, \tag{2.2}$$

$$s_{rj}(h; p) = \frac{e^{V_{rj}(h;p)}}{Z_r(h; p)}, \qquad s_{r0}(h; p) = \frac{1}{Z_r(h; p)}, \tag{2.3}$$

$$s_{r0} + s_{r1} + s_{r2} = \frac{1 + e^{V_{r1}} + e^{V_{r2}}}{Z_r} = 1. \tag{2.4}$$

Suppress history and prices temporarily. For a price p_j,

$$\frac{\partial V_{rk}}{\partial p_j} = \alpha \mathbb{1}\{k = j\}, \tag{2.5}$$

$$\frac{\partial e^{V_{rk}}}{\partial p_j} = e^{V_{rk}} \frac{\partial V_{rk}}{\partial p_j} = \alpha e^{V_{rk}} \mathbb{1}\{k = j\}, \tag{2.6}$$

$$\frac{\partial Z_r}{\partial p_j} = \sum_{l=1}^{2} \alpha e^{V_{rl}} \mathbb{1}\{l = j\} = \alpha e^{V_{rj}}, \tag{2.7}$$

$$\frac{\partial s_{rk}}{\partial p_j} = \frac{Z_r \alpha e^{V_{rk}} \mathbb{1}\{k = j\} - e^{V_{rk}} \alpha e^{V_{rj}}}{Z_r^2} \tag{2.8}$$

$$= \alpha \frac{e^{V_{rk}}}{Z_r} \mathbb{1}\{k = j\} - \alpha \frac{e^{V_{rk}}}{Z_r} \frac{e^{V_{rj}}}{Z_r} \tag{2.9}$$

$$= \alpha s_{rk} (\mathbb{1}\{k = j\} - s_{rj}). \tag{2.10}$$

Thus the own and cross elasticities are, respectively,

$$\epsilon_{rjj} = \frac{p_j}{s_{rj}} \frac{\partial s_{rj}}{\partial p_j} = \frac{p_j}{s_{rj}} \alpha s_{rj}(1 - s_{rj}) = \alpha p_j (1 - s_{rj}), \tag{2.11}$$

$$\epsilon_{rkj} = \frac{p_j}{s_{rk}} \frac{\partial s_{rk}}{\partial p_j} = \frac{p_j}{s_{rk}} (-\alpha s_{rk} s_{rj}) = -\alpha p_j s_{rj}, \qquad k \neq j. \tag{2.12}$$

First-period utility and shares have no purchase-history state:

$$V_{rj1} = \beta_j + \alpha p_{j1}, \tag{2.13}$$

$$S_{rj}(p_1) = \frac{e^{\beta_j + \alpha p_{j1}}}{1 + \sum_k e^{\beta_k + \alpha p_{k1}}}, \tag{2.14}$$

$$S_{Hj} = S_{Lj}, \tag{2.15}$$

$$D_{r,kj}(p_1) := \frac{\partial S_{rk}}{\partial p_{j1}} = \alpha S_{rk}(\mathbb{1}\{k = j\} - S_{rj}). \tag{2.16}$$

The first purchase has the same demand for both types. Thus purchase masses $\omega_r S_{rh}$ factor into the known type mass and a common purchase rate.

3. Common prices chosen using the known masses

Write $p_{k2}(h)$ for the terminal price of product k following product h. For any proposed menu, define

$$m_{j1} = p_{j1} - c_j, \qquad m_{k2}(h) = p_{k2}(h) - c_k, \tag{3.1}$$

$$w_r(h; p_2(h)) = \sum_{k=1}^{2} s_{rk}(h; p_2(h)) m_{k2}(h), \tag{3.2}$$

$$A_{rh} = m_{h1} + \delta w_r(h; p_2(h)). \tag{3.3}$$

The expected mass of type-r consumers purchasing h is $N\omega_r S_{rh}$. This is a joint purchase mass, computed before purchases, not an updated type probability. Expected profits follow by multiplying each mass by its associated margin:

$$\pi_1 = N \sum_r \sum_h \omega_r S_{rh} m_{h1}, \tag{3.4}$$

$$\pi_2 = N \sum_r \sum_h \sum_k \omega_r S_{rh} s_{rk}(h; p_2(h)) m_{k2}(h), \tag{3.5}$$

$$\Pi = \pi_1 + \delta \pi_2 \tag{3.6}$$

$$= N \sum_r \sum_h \omega_r S_{rh} m_{h1} + N\delta \sum_r \sum_h \omega_r S_{rh} \sum_k s_{rk}(h; p_2(h)) m_{k2}(h) \tag{3.7}$$

$$= N \sum_r \omega_r \sum_h S_{rh} [m_{h1} + \delta w_r(h; p_2(h))] \tag{3.8}$$

$$= N \sum_r \omega_r \sum_h S_{rh} A_{rh}. \tag{3.9}$$

The common-menu problem is

$$\max_{p_1, p_2(1), p_2(2) \in [1,8]^2} \Pi.$$

All six prices are selected together. "Common" means across types; products, periods, and purchase histories may have different prices.

3.1. First-period derivatives

For a partial derivative, hold the other menu entries fixed. In particular, $w_r(h; p_2(h))$ has no direct dependence on p_{j1}. Therefore

$$\frac{\partial m_{h1}}{\partial p_{j1}} = \mathbb{1}\{h = j\}, \tag{3.10}$$

$$\frac{\partial A_{rh}}{\partial p_{j1}} = \mathbb{1}\{h = j\} + \delta\, 0 = \mathbb{1}\{h = j\}, \tag{3.11}$$

$$\frac{1}{N}\frac{\partial \Pi}{\partial p_{j1}} = \sum_r \omega_r \sum_h \frac{\partial (S_{rh} A_{rh})}{\partial p_{j1}} \tag{3.12}$$

$$= \sum_r \omega_r \sum_h \left[\frac{\partial S_{rh}}{\partial p_{j1}} A_{rh} + S_{rh} \frac{\partial A_{rh}}{\partial p_{j1}} \right] \tag{3.13}$$

$$= \sum_r \omega_r \left[\sum_h D_{r,hj} A_{rh} + \sum_h S_{rh} \mathbb{1}\{h = j\} \right] \tag{3.14}$$

$$= \sum_r \omega_r \left[S_{rj} + \sum_h D_{r,hj} (m_{h1} + \delta w_r(h; p_2(h))) \right]. \tag{3.15}$$

At an interior optimum this equals zero. With two types it is explicitly

$$\begin{aligned} 0 = {} & \lambda \left[S_{Hj} + D_{H,1j} A_{H1} + D_{H,2j} A_{H2} \right] \\ & + (1 - \lambda) \left[S_{Lj} + D_{L,1j} A_{L1} + D_{L,2j} A_{L2} \right]. \end{aligned} \tag{3.16}$$

No derivative of a type probability appears: λ is fixed.

3.2. Second-period derivatives

For a given history h, first differentiate conditional terminal profit:

$$\frac{\partial w_r(h)}{\partial p_{j2}(h)} = \sum_k \left[\frac{\partial s_{rk}(h)}{\partial p_{j2}(h)} m_{k2}(h) + s_{rk}(h) \mathbb{1}\{k = j\} \right] \tag{3.17}$$

$$= \sum_k \alpha s_{rk}(h) (\mathbb{1}\{k = j\} - s_{rj}(h)) m_{k2}(h) + s_{rj}(h) \tag{3.18}$$

$$= \alpha s_{rj}(h) m_{j2}(h) - \alpha s_{rj}(h) \sum_k s_{rk}(h) m_{k2}(h) + s_{rj}(h) \tag{3.19}$$

$$= s_{rj}(h) \left[1 + \alpha (m_{j2}(h) - w_r(h)) \right]. \tag{3.20}$$

Only continuation at history h uses the entry $p_{j2}(h)$, so

$$\frac{1}{N}\frac{\partial \Pi}{\partial p_{j2}(h)} = \delta \sum_r \omega_r S_{rh} \frac{\partial w_r(h)}{\partial p_{j2}(h)} \tag{3.21}$$

$$= \delta \sum_r \omega_r S_{rh} s_{rj}(h)[1 + \alpha(m_{j2}(h) - w_r(h))] \tag{3.22}$$

$$= \delta\{\lambda S_{Hh} s_{Hj}(h)[1 + \alpha(m_{j2}(h) - w_H(h))] + (1-\lambda) S_{Lh} s_{Lj}(h)[1 + \alpha(m_{j2}(h) - w_L(h))]\}. \tag{3.23}$$

Since $S_{Hh} = S_{Lh}$, the common first-period purchase rate factors out of each terminal condition. Terminal prices therefore maximize $\sum_r \omega_r w_r(h)$ directly using the known type masses. Equal terminal markups across goods need not hold for a mixture of types.

For any price coordinate x, write $g_x = \partial \Pi / \partial x$. The necessary bound conditions for maximization are

$$g_x = 0 \text{ if } 1 < x < 8, \qquad g_x \leq 0 \text{ if } x = 1, \qquad g_x \geq 0 \text{ if } x = 8.$$

At the lower bound only increases are feasible, so a positive derivative would improve profit; at the upper bound only decreases are feasible, so a negative derivative would improve profit. These conditions are necessary, not a proof of a global maximum.

4. Observed-type benchmark and price identities

If type is observed exogenously, give each type a separate menu. Then

$$\Pi^{\text{obs}} = N \sum_r \omega_r \Pi_r, \tag{4.1}$$

$$\Pi_r = \sum_h S_{rh}[p_{rh1} - c_h + \delta w_r(h)], \tag{4.2}$$

$$w_r(h) = \max_{p_{r2}(h) \in [1,8]^2} \sum_k s_{rk}(h; p_{r2}(h))[p_{rk2}(h) - c_k]. \tag{4.3}$$

For $\omega_r > 0$, multiplying Π_r by $N\omega_r$ does not change its maximizing menu, because it multiplies every profit comparison by a positive constant. At zero mass its menu is irrelevant to aggregate profit; the code still computes it as a benchmark. These observed-type menus have no learning. Terminal prices can be solved first, followed by the initial prices.

The initial derivative is the common-price derivative with a single type:

$$0 = S_{rj} + \sum_h D_{r,hj}[m_{rh1} + \delta w_r(h)].$$

Write $s_{rk1} = S_{rk}$ and $a_{rk} = m_{rk1} + \delta w_r(k)$. Substitution and division by $s_{rj1} > 0$ give

$$0 = s_{rj1} + \sum_k \alpha s_{rk1}(\mathbb{1}\{k = j\} - s_{rj1})a_{rk} \tag{4.4}$$

$$= s_{rj1} + \alpha s_{rj1} a_{rj} - \alpha s_{rj1} \sum_k s_{rk1} a_{rk}, \tag{4.5}$$

$$0 = 1 + \alpha \left[a_{rj} - \sum_k s_{rk1} a_{rk} \right]. \tag{4.6}$$

To display the outside option explicitly,

$$a_{rj} - \sum_k s_{rk1} a_{rk} = (1 - s_{rj1}) a_{rj} - \sum_{k \neq j} s_{rk1} a_{rk} \tag{4.7}$$

$$= \left(s_{r01} + \sum_{k \neq j} s_{rk1} \right) a_{rj} - \sum_{k \neq j} s_{rk1} a_{rk} \tag{4.8}$$

$$= s_{r01} a_{rj} + \sum_{k \neq j} s_{rk1}(a_{rj} - a_{rk}). \tag{4.9}$$

Solving the first-order equation algebraically,

$$-1 = \alpha \left[a_{rj} - \sum_k s_{rk1} a_{rk} \right], \tag{4.10}$$

$$-\frac{1}{\alpha} = a_{rj} - \sum_k s_{rk1} a_{rk}, \tag{4.11}$$

$$a_{rj} = -\frac{1}{\alpha} + \sum_k s_{rk1} a_{rk}, \tag{4.12}$$

$$p_{rj1} - c_j + \delta w_r(j) = -\frac{1}{\alpha} + \sum_k s_{rk1}[p_{rk1} - c_k + \delta w_r(k)], \tag{4.13}$$

$$p_{rj1} = c_j - \delta w_r(j) - \frac{1}{\alpha} + \sum_k s_{rk1}[p_{rk1} - c_k + \delta w_r(k)]. \tag{4.14}$$

Subtracting the equations for j and l cancels the common sum:

$$a_{rj} - a_{rl} = 0, \tag{4.15}$$

$$(p_{rj1} - c_j + \delta w_r(j)) - (p_{rl1} - c_l + \delta w_r(l)) = 0, \tag{4.16}$$

$$p_{rj1} - p_{rl1} = c_j - c_l - \delta[w_r(j) - w_r(l)]. \tag{4.17}$$

These identities are implicit, since shares depend on prices, and this relative-price identity assumes interior prices.

For a single observed type at terminal history h, the terminal derivative already derived

gives

$$0 = s_{rj2}(h)[1 + \alpha(m_{rj2}(h) - w_r(h))], \tag{4.18}$$

$$0 = 1 + \alpha(m_{rj2}(h) - w_r(h)), \tag{4.19}$$

$$m_{rj2}(h) = w_r(h) - 1/\alpha. \tag{4.20}$$

The right side is the same for every j, so all inside margins equal $M_r(h)$. Consequently

$$w_r(h) = \sum_k s_{rk2}(h) M_r(h) = (1 - s_{r02}(h)) M_r(h), \tag{4.21}$$

$$M_r(h) = (1 - s_{r02}(h)) M_r(h) - 1/\alpha, \tag{4.22}$$

$$M_r(h) - (1 - s_{r02}(h)) M_r(h) = -1/\alpha, \tag{4.23}$$

$$s_{r02}(h) M_r(h) = -1/\alpha, \tag{4.24}$$

$$M_r(h) = -\frac{1}{\alpha s_{r02}(h)}, \tag{4.25}$$

$$p_{rj2}(h) - p_{rl2}(h) = [c_j + M_r(h)] - [c_l + M_r(h)] = c_j - c_l. \tag{4.26}$$

The outside share must be retained. These equalities assume interior prices.

5. Pricing directly from a known two-point distribution

All consumers begin with no previous product. Write $\rho = \Pr(L) = 1 - \lambda$, so $\Pr(H) = 1 - \rho$. There is no first-period switching bonus:

$$V^B_{rj1} = \beta_j + \alpha p_{j1}, \tag{5.1}$$

$$b_j(p_1) = \frac{\exp(\beta_j + \alpha p_{j1})}{1 + \sum_k \exp(\beta_k + \alpha p_{k1})}, \tag{5.2}$$

$$S^B_{Hj} = S^B_{Lj} = b_j(p_1). \tag{5.3}$$

Known type masses are used directly in terminal expected profits:

$$\bar{w}(h; p) = (1 - \rho) w_H(h; p) + \rho w_L(h; p), \tag{5.4}$$

$$p^*_2(h; \rho) \in \arg\max_{p \in [1,8]^2} \bar{w}(h; p), \tag{5.5}$$

$$W_h(\rho) = \bar{w}(h; p^*_2(h; \rho)). \tag{5.6}$$

The terminal interior condition is explicitly

$$0 = \frac{\partial \bar{w}(h;p)}{\partial p_j} = (1-\rho)s_{Hj}(h;p)[1+\alpha(p_j - c_j - w_H(h;p))] + \rho s_{Lj}(h;p)[1+\alpha(p_j - c_j - w_L(h;p))]. \tag{5.7}$$

Since first-period shares are identical across types, the total objective factors without any conditioning on type:

$$\Pi^B/N = (1-\rho)\sum_h b_h[m_{h1} + \delta w_H(h)] + \rho \sum_h b_h[m_{h1} + \delta w_L(h)] \tag{5.8}$$

$$= \sum_h b_h[(1-\rho)m_{h1} + \rho m_{h1} + \delta((1-\rho)w_H(h) + \rho w_L(h))] \tag{5.9}$$

$$= \sum_h b_h[m_{h1} + \delta \bar{w}(h)] \tag{5.10}$$

$$= \sum_h b_h[p_{h1} - c_h + \delta W_h(\rho)] \quad \text{at the optimal terminal menu.} \tag{5.11}$$

Because $W_h(\rho)$ has no dependence on p_1, terminal prices can be chosen first. Put $A_h^B = p_{h1} - c_h + \delta W_h(\rho)$. Then

$$\frac{1}{N}\frac{\partial \Pi^B}{\partial p_{j1}} = \sum_h \left[\alpha b_h(\mathbb{1}\{h=j\} - b_j)A_h^B + b_h \mathbb{1}\{h=j\}\right] \tag{5.12}$$

$$= \alpha b_j A_j^B - \alpha b_j \sum_h b_h A_h^B + b_j \tag{5.13}$$

$$= b_j\left[1 + \alpha\left(A_j^B - \sum_h b_h A_h^B\right)\right]. \tag{5.14}$$

For an interior solution, divide by $b_j > 0$ and rearrange:

$$0 = 1 + \alpha\left(A_j^B - \sum_h b_h A_h^B\right), \tag{5.15}$$

$$A_j^B = -1/\alpha + \sum_h b_h A_h^B, \tag{5.16}$$

$$p_{j1} = c_j - \delta W_j(\rho) - 1/\alpha + \sum_h b_h[p_{h1} - c_h + \delta W_h(\rho)]. \tag{5.17}$$

This is pricing under a known probability mass function, with no learning.

5.1. A fixed discount decision and every cutoff step

Construct a regular menu R from the known-H newborn optimum. Construct menu D by holding its initial prices fixed and replacing terminal prices with the known-L terminal optimum. The code verifies that these terminal prices are weakly lower. The full known-L dynamic policy is not necessarily a discount: its introductory prices can be higher because the H consumer has greater continuation value. Both comparison menus stay fixed as ρ varies.

For $X \in \{R, D\}$, define profits conditional on true type and the profit change from discounting:

$$F_r^X = \sum_h b_h(p_1^R)[p_{h1}^R - c_h + \delta w_r(h; p_2^X(h))], \tag{5.18}$$

$$\Delta_r = F_r^D - F_r^R \tag{5.19}$$

$$= \sum_h b_h(p_1^R)\{[p_{h1}^R - c_h + \delta w_r(h; p_2^D(h))] - [p_{h1}^R - c_h + \delta w_r(h; p_2^R(h))]\} \tag{5.20}$$

$$= \delta \sum_h b_h(p_1^R)[w_r(h; p_2^D(h)) - w_r(h; p_2^R(h))]. \tag{5.21}$$

By the known type probabilities, the expected gain is

$$B(\rho) = [(1-\rho)F_H^D + \rho F_L^D] - [(1-\rho)F_H^R + \rho F_L^R] \tag{5.22}$$

$$= (1-\rho)(F_H^D - F_H^R) + \rho(F_L^D - F_L^R) \tag{5.23}$$

$$= (1-\rho)\Delta_H + \rho\Delta_L \tag{5.24}$$

$$= \Delta_H - \rho\Delta_H + \rho\Delta_L \tag{5.25}$$

$$= \Delta_H + \rho(\Delta_L - \Delta_H). \tag{5.26}$$

For $\Delta_H < 0 < \Delta_L$, the slope is positive, and the indifference point satisfies

$$0 = \Delta_H + \rho^*(\Delta_L - \Delta_H), \tag{5.27}$$

$$-\Delta_H = \rho^*(\Delta_L - \Delta_H), \tag{5.28}$$

$$\rho^* = \frac{-\Delta_H}{\Delta_L - \Delta_H}. \tag{5.29}$$

Because the denominator is positive, the discount rule is obtained without reversing the

inequality:

$$B(\rho) > 0 \iff \Delta_H + \rho(\Delta_L - \Delta_H) > 0 \tag{5.30}$$

$$\iff \rho(\Delta_L - \Delta_H) > -\Delta_H \tag{5.31}$$

$$\iff \rho > \frac{-\Delta_H}{\Delta_L - \Delta_H} = \rho^*. \tag{5.32}$$

For these menus the numerical values are approximately

$$\Delta_H = -0.04484198, \qquad \Delta_L = 0.02410654, \tag{5.33}$$

$$\Delta_L - \Delta_H = 0.02410654 - (-0.04484198) = 0.06894852, \tag{5.34}$$

$$\rho^* \simeq \frac{0.04484198}{0.06894852} = 0.650369, \tag{5.35}$$

$$B(1/2) \simeq \tfrac{1}{2}(-0.04484198) + \tfrac{1}{2}(0.02410654) \tag{5.36}$$

$$= -0.02242099 + 0.01205327 = -0.01036772, \tag{5.37}$$

$$1 - \rho^* \simeq 1 - 0.650369 = 0.349631. \tag{5.38}$$

Offer the fixed discount above approximately 65.04% probability of L; below it retain the regular menu; at equality either menu is optimal. At a 50/50 mass the discount loses profit. The cost of discounting an H type is included in Δ_H and requires no extra fee.

If $a \in [0, 1]$ instead denotes the probability of sending the fixed discount,

$$F(a; \rho) = (1 - a)F^R(\rho) + aF^D(\rho) \tag{5.39}$$

$$= F^R(\rho) - aF^R(\rho) + aF^D(\rho) \tag{5.40}$$

$$= F^R(\rho) + aB(\rho), \tag{5.41}$$

$$\frac{\partial F(a; \rho)}{\partial a} = B(\rho). \tag{5.42}$$

Thus $a = 1$ maximizes profit when $B > 0$, $a = 0$ when $B < 0$, and every a is optimal when $B = 0$. This is distinct from varying the type mass ρ. A freely optimized common menu weakly dominates either fixed menu because both fixed menus are feasible choices in its optimization problem.

6. Numerical implementation and profit distribution

The parameters are

$$\beta = (2.2, 2.0), \quad c = (1.2, 1.0), \quad \alpha = -1.5, \quad \delta = 0.95, \quad \gamma_H = 2, \quad \gamma_L = 0.1.$$

Run `python3 toy_model.py`. The common-menu solver jointly chooses all prices using the fixed mass objective above. Terminal problems can also be solved first because first-period demand is identical across types. Optimization uses three starting values and checked solver status; this is not a global optimality proof. The fixed terminal-discount comparison uses the same model and holds the regular first-period prices fixed.

The results directory contains `prices_50_50.csv`, `population_mix.csv`, `profit_by_gamma.csv`, `newborn_prices.csv`, and `newborn_discount_probability.csv`, together with reports and figures. The newborn CSV labels ρ as `probability_L`. The population experiment varies $\lambda = \Pr(H)$; the newborn experiment varies $\rho = \Pr(L)$.

For any fixed common policy P, let $f_r(P)$ be expected discounted profit conditional on type. The distribution of these conditional expectations has two masses, with cumulative distribution function

$$\Pr(f_r(P) \leq x) = \lambda \mathbb{1}\{f_H(P) \leq x\} + (1-\lambda)\mathbb{1}\{f_L(P) \leq x\}.$$

Its expectation is

$$\mathbb{E}[f_r(P)] = \lambda f_H(P) + (1-\lambda) f_L(P).$$

If both profit values coincide, their masses add to one at that value. This describes type-conditional expected profits, rather than realized transaction profits. A curve of expected profit against γ is not a probability density; a continuous type distribution would have to be specified to construct a continuous distribution of conditional expected profits.